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# Design and Implementation of a Hands-Free Electrolarynx Device Controlled by Neck Strap Muscle Electromyographic Activity

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*Abstract* - **Total laryngectomy surgery results in a loss of human voice, and a corresponding deficit for normal speech production. A key disadvantage of available voice replacement prostheses is the inconvenience of using one hand for operating the device when speaking. To address this problem, a hands-free portable electrolarynx (EL) was developed. This wearable device transforms neck muscle electromyographic (EMG) activity into control signals for initiation and termination of voicing. Using the new system (EMG-EL), seven normal subjects produced voice initiation that was as fast as with a commercial hand-held EL, and faster than with normal voice. The EMG-EL was also tested in a laryngectomy subject, who produced voice initiation that was slower than with the commercial EL, but faster than with tracheo-esophageal voice. Voice termination with the EMG-EL was slower than with the other voice sources for all subjects.**

*Keywords* - **Electromyography, voice, prosthesis**

## I. Introduction

Human voice is the foundation of self-expression and vocal communication with others. Unfortunately, each year thousands of people undergo a laryngectomy, which is the surgical removal of the larynx for the treatment of cancer. Loss of the larynx results in an inability to produce normal speech and in the need to breathe through a hole (stoma) in the neck.

However, since the main articulators are still intact, a prosthetic device can be used to acoustically excite the vocal tract for production of alaryngeal speech. The majority of laryngectomy patients either use a tracheo-esophageal (TE) prosthesis, or an electrolarynx (EL). A TE prosthesis is a one-way valve that diverts air from the trachea into the esophagus to produce a belch-like voice. The EL is a hand-held battery-operated buzzer that injects acoustic energy into the throat or mouth. Use of the EL requires little training and produces intelligible speech, making it a popular voice aid. Multiple studies report that over half of the laryngectomy patients use an EL for verbal communication [1-3].

A major inconvenience in using a conventional TE prosthesis or a hand-held EL device is that they occupy the use of one hand. Before being able to speak or respond to a question, a TE user usually has to manually occlude the stoma in the neck. Similarly, an EL user has to reach into his pocket for the device and apply it to the neck surface or oral cavity. A recent survey of EL users found that the inconvenience of use was ranked most problematic, followed by the monotonic nature of the speech [4]. These results indicated the potential benefit in providing hands-free EL control.

A hand-held EL is composed of three parts: a transducer that mechanically produces the sound, an electric circuit that drives the transducer, and a battery pack that powers the device. All three parts are usually contained in one unit that delivers sound into the neck or the oral cavity. On/off control is typically achieved using a button switch.

The system described here uses electromyographic (EMG) signals for on/off control, following the approach adopted in the fields of limb prostheses [5-9], and human-computer interface technology [10-12]. Electrical signals from neck muscles are detected using a surface electrode, and are processed to produce on/off control. This report covers device design and assessment of system performance relative to normal, EL, and TE alaryngeal voice.

## II. Methods

### A. The EMG-EL System

The EMG processing circuit of the EMG-controlled EL (EMG-EL) produces an envelope waveform proportional to the time-averaged power in the EMG signal. The time-varying envelope controls on/off triggering and fundamental frequency (pitch) of a commercial EL transducer held against the neck with a simple brace. A schematic representation of the system is shown in Fig. 1.

The electrical activity of the neck strap muscles is detected on the surface of the skin using a bipolar electrode (Delsys Inc., Boston MA). The EMG signal amplitude is on the order of tens of micro-volts, with most energy in the

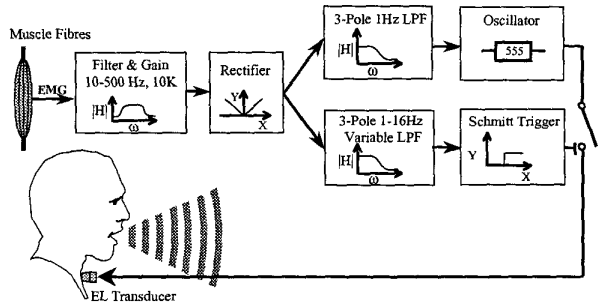


Fig. 1: Processing stages of the EMG-EL processor. EMG signal is filtered and amplified, then rectified and smoothed to produce a slow envelope (top path) for pitch control, and a fast envelope (bottom path) for on/off control of the EL.

frequency range from 10 to 500Hz. Therefore, the EMG signal was amplified and band-pass filtered to increase signal-to-noise ratio. The signal was then actively rectified and sent through two parallel pathways, each with a 3-pole low-pass filter (LPF). The first LPF has a corner frequency of approximately 1Hz, while the second has a variable corner frequency that could be adjusted from 1Hz to 16Hz. The two pathways produce two envelopes with 'slow' and 'fast' time constants.

The fast envelope is fed to a Schmitt trigger to turn the transducer on and off based on a controllable threshold voltage. Simultaneously, the slow envelope is used to control the fundamental frequency of EL voice by directly modifying the frequency of the oscillator driving an EL transducer (NuVois, Mountain Precision Mfg., ID). The other parameters of the EMG-EL processor are EMG gain, EL volume, starting fundamental frequency, and corner frequency for the fast-varying envelope. This paper will discuss voice onset and termination, but not control of fundamental frequency .

### *B. EMG-EL Performance Assessment*

To compare user performance with the EMG-EL to other voice sources, a reaction time protocol was adopted [13-16]. A visual cue signaled the subject to start and stop vocalizing, and the acoustic output was recorded. The time delay between the presentation of the visual cue and voice initiation and termination was used as a measure of vocal reaction time. Voice initiation time (VIT) and voice termination time (VTT) were measured for the EMG-EL, a conventional button EL, and normal voice in seven normal subjects, as well as TE voice in one laryngectomy subject.

The normal subjects were 4 males and 3 females, whose ages ranged from 21 to 49 years, averaging 27.5 years. The laryngectomy patient was a 61 year old male who underwent a total laryngectomy 30 months prior to testing. The omohyoid muscles were left intact after resection of the larynx in this subject. The right omohyoid retained its original innervation, while the left omohyoid nerve was connected to the left recurrent laryngeal nerve. The laryngectomy subject initially used an EL, until he was fitted with a TE valve a few months after the surgery. The subject primarily used TE speech, but also used the EL on occasion whenever his TE valve was not functional.

Each subject was seated in a sound-treated chamber, with a video monitor (Sony Trinitron GVM 2020) placed one meter away. Speech signals were recorded with a condenser microphone (Sony ECM-50PSW) 15cm from the mouth. The stimulus presentation, data acquisition and analysis were performed using a PC. Matlab (Mathworks, Natick, MA) software controlled the sound card (Aureal Semiconductor, Vortex AU8830 PCI) as a 2-channel input device. For timing accuracy, a photocell was mounted on the video monitor to detect the instant of visual stimulus presentation. The photocell output was recorded along with speech signals. Using the acoustic signal from the microphone and the electric signal from the photocell, the time delays between the stimulus and voice initiation and termination were measured with a 1ms resolution (Fig. 2).

EMG signals were detected with a non-invasive DelSys DE2.1 surface electrode placed over the neck strap muscles. The EMG electrode was positioned along the midline of the neck, directly below the thyroid prominence of the normal subjects, and over the left omohyoid muscle above the stoma of the laryngectomy subject. An Ag/AgCl gel electrode was placed on the shoulder.

Normal subjects received brief training on how to use the EL (Servox INTON, Cologne, Germany) and the EMG-EL, while the laryngectomy subject underwent a more extensive training protocol. The EMG gain was set so that the peak EMG envelope elicited by a low pitch vocalization was about 500mV. The electrode contact with the skin was adjusted until the resting baseline was less than 50mV. If the baseline was any larger than 50mV, the ground and EMG electrode connections were inspected for faulty coupling with the skin surface. Conductive electrode gel was used when necessary. The EMG-EL threshold determined whether the sound transducer was on or off, based on the amplitude of the fast EMG envelope. The threshold was set at the baseline level + 10% of the observed range in envelope amplitude. For example, if the baseline was 30mV and the maximum EMG envelope was 500mV, then the threshold was set at 30mV + (500-30) mV/10 = 77mV.

Within each trial, a sequence of visual commands with different background colors was displayed on the video monitor. The bottom panel of Fig. 2 shows a schematic of the sequence of events in a trial. Each trial started with a rest period of 10 seconds, followed by a 'get ready' period of 1 or 2 seconds. The variable ready period was chosen to reduce the effect of the subject's anticipation and to minimize the reaction time [16]. The ready period was

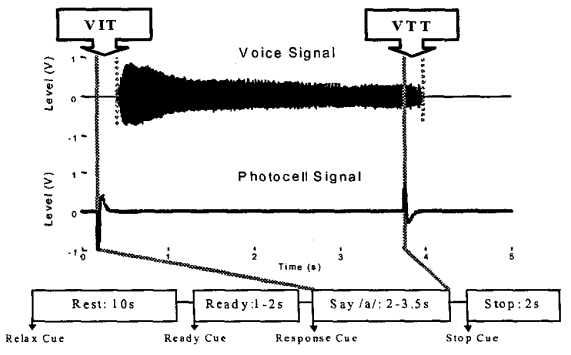


Fig. 2: Visual stimulus for the voice initiation and termination time experiment. Top trace shows audio signal of subject voice, while bottom trace shows photocell signal indicating visual stimulus timing.

followed by the 'response' cue, where the display showed the command 'Say /a/' on a bright green background. The response interval was 2, 2.5, 3, or 3.5 seconds. Ending the response period was the stop signal, which was a display of the word 'Stop' on a red background for 2 seconds. The stimulus sequence was repeated until all trials in a run were completed.

A run included 8 trials, each with a pseudo-randomized ready and response period length combination. Each voice source (Button-EL, EMG-EL, TE and normal voice) was tested for 5 runs. A run was repeated only in the infrequent event of failure to maintain voicing during the response period. The order of testing of the different voice sources was also randomized for each of the normal subjects. At the end of the experiment there were 5 runs × 8 trials = 40 tokens per voice source per subject. The different voice sources were treated as the repeated measure within each subject. The normal subject group therefore had a total of 280 tokens per voice source, while the laryngectomy subject had 40 tokens per voice source.

Before submitting the data to an analysis of variance (ANOVA), the reaction times were converted into reaction rates by taking the reciprocal of each data point. The reciprocals of VIT and VTT produced the voice initiation rate (VIR) and voice termination rate (VTR). This variable transformation yielded distributions that were much less skewed and more normal, and therefore more appropriate for the ANOVA. For example, the average skewness for the VIT variables of normal subjects was 2.08, and dropped to 0.37 for the VIR. Similarly, for laryngectomy voice initiation, the skewness for VIT was 1.94 on average, and dropped to 0.36 for the VIR following data transformation.

## III. RESULTS

### A. *The EMG-EL System*

The EMG-EL processor prototype was built and optimized for battery power consumption in collaboration with Draper Labs (Cambridge, MA). The dimensions of the processing unit are 5cm×15cm×17.5cm, with an 1/8 in. connector for the EL transducer, a hypertronics 4-pole connector for the DelSys EMG electrode, and 5 knobs for threshold, EMG gain, starting fundamental frequency, transducer volume, and LPF corner frequency (Fig. 3a).

The DE2.1 DelSys surface EMG electrode was 2cm×4.1cm×0.5cm in size, with two silver bars 1cm long and 0.1cm wide, spaced 1cm apart. The electrode required a supply voltage of ±4.5Volts, with 2.5mA maximum supply current, and provided a head gain of 10±5%. The transducer used in conjunction with the EMG-EL system was identical to that used in NuVois EL devices. It contained a magnet, coil-and-plunger, and drum assembly. It was mounted on a metal brace made of a thick copper wire covered within plastic tubing for comfort. The brace

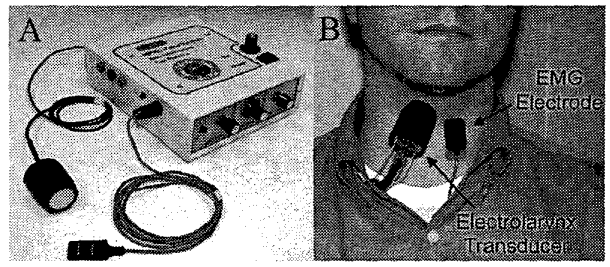


Fig. 3: The EMG-EL System. A: DelSys EMG electrode, EMG-EL processor, and EL transducer. B: The transducer and EMG electrode are worn on the neck for hands-free EL speech.

was inconspicuously wrapped around the back of the neck and down the middle of the chest for support (Fig. 3b).

The processor box containing the battery pack and electrical circuit was mounted on the user's belt. The physical design of the EMG-EL processor was largely determined by the batteries (9V type for circuit and six AA type for transducer). The EMG electrode and processing circuit were composed of low-power analog components that required an average of 5mA of current. Given that an alkaline 9V battery typically has 500mAh capacity [17], the electrode and processor could operate for 100 hours before requiring battery replacement.

The transducer power requirement is much larger than that of the control circuit, averaging a current load of 450mA. If we assume the capacity of each AA battery to be 1000mAh [17], then the 6 AA alkaline batteries were estimated to last for about 13 hours of continuous operation.

### B. *EMG-EL Performance Assessment*

Fig. 4 shows the VIR and VTR data averages for all subjects. The VIR of normal subjects was significantly different across voice sources ($p<0.001$, $F=26.5$, $N=280$). Post-hoc tests showed that the 3.8 $s^{-1}$ VIR of normal voice was significantly slower than the 4.21 $s^{-1}$ VIR for the button-EL, and the 4.11 $s^{-1}$ VIR for the EMG-EL ($p<0.001$), but button-EL and EMG-EL were not significantly different from each other. For termination, the VTR averages across voice sources also showed significant differences ($p<0.001$, $F=8.3$, $N=280$). Post-hoc tests showed that the 3.86 $s^{-1}$ VTR

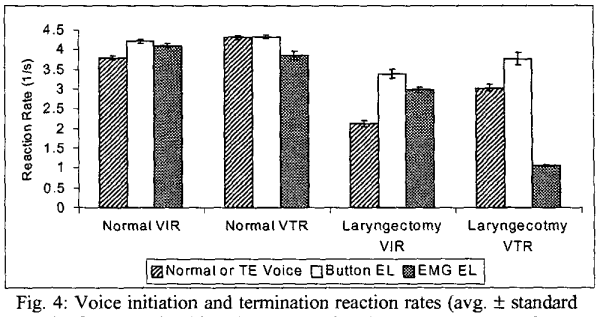


Fig. 4: Voice initiation and termination reaction rates (avg. ± standard error) of 7 normal subjects' normal voice, button-EL, and EMG-EL, and a laryngectomy subject using TE voice, button-EL, and EMG-EL.

of the EMG-EL was significantly slower than the 4.3 $s^{-1}$ VTR of normal voice and the 4.31 $s^{-1}$ VTR of the button-EL ($p<0.001$).

In the case of the laryngectomy subject, analysis of variance showed that the differences between the average values of VIR for the different voicing modes was significant ($p<0.001$, $F=55$, $N=40$). VTR data for the different voice sources also showed a similar outcome ($p<0.001$, $F=338$, $N=40$). Post-hoc tests showed that all pair-wise differences were significant for the various voice sources used by the laryngectomy subject. The 2.97 $s^{-1}$ VIR of the EMG-EL fell between the faster 3.39 $s^{-1}$ VIR of the button-EL and the slower 2.12 $s^{-1}$ VIR of TE voice. EMG-EL VTR, on the other hand, was 1.07 $s^{-1}$, which was slower than both the 3.77 $s^{-1}$ VTR of the button-EL and 3.03 $s^{-1}$ VTR of TE voice.

Although the laryngectomy subject had reaction rates for all three voice sources that were generally slower compared to the normal speakers, the pattern of his results was similar to normal subjects. For example, EMG-EL voice initiation was faster than normal voice in normal subjects, and TE voice in the laryngectomy subject. In addition, EMG-EL voice initiation was slower than button-EL for both subject types. Furthermore, EMG-EL voice termination was clearly slowest for both laryngectomy and normal subjects.

## IV. DISCUSSION

The prototype EMG-EL device is based on a simple processing scheme using analog components. Although the application of digital circuits for more sophisticated processing could potentially improve the performance of the device, the current method of processing appears to be sufficient as a proof of concept for using EMG signals to control an EL.

The VIR results showed that the EMG-EL is as fast as a commercial EL, and faster than both normal and TE in voice initiation. The VTR results showed that EMG-EL voice termination is not as fast as the other voice sources tested. Slow VTR of the EMG-EL might be explained by the fact that it depends on strap muscle relaxation, which is not an active process. Normal voice termination is achieved either by the active contraction of the laryngeal adductor muscles to completely close the glottis, or the action of the abductor muscles to pull the vocal folds apart. TE and button-EL voice termination was achieved by the active process of lifting the finger away from either the stoma or the button-switch. The EMG-EL lacked a corresponding set of muscles that could actively terminate voicing.

Furthermore, the low pass filtering of the EMG signal and the relatively low threshold setting of 10% above the baseline produced fast voice initiation rates at the expense of slow voice termination. This issue could be addressed through the use of different thresholds for voice initiation and termination. In addition, training and practice might yield better conscious control over the neck muscles triggering the EMG-EL. Although subjects were unable to produce VTR comparable to that obtained with the other voice sources, this result might not have a serious impact when the EMG-EL is used for speech. During speech production, the user can possibly anticipate when pauses will occur, and adjust the activity of the muscles controlling the device to produce timely voice termination.

Finally, the VIR and VTR are only two of many metrics that could be used to evaluate the efficacy of EMG-EL speech. Future work will be directed at testing the EMG-EL in the production of words, sentences, and paragraphs. In addition, preliminary observations indicate that natural-sounding intonation can be readily incorporated into EMG-EL voice by using the slow EMG envelope to modulate the fundamental frequency.

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